2.1 Equilibrium

Learning Goals (Students will be able too…)

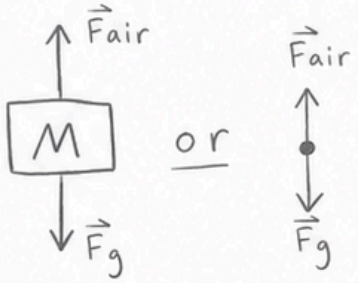
- Be able to solve force problems when forces are acting in two dimensions.
- Being able to distinguish between static and dynamic equilibrium.
- Draw FBD diagrams depicting all the forces acting on an object.
- Calculate the equilibrant to a set of forces.

Free Body Diagrams (FBD)

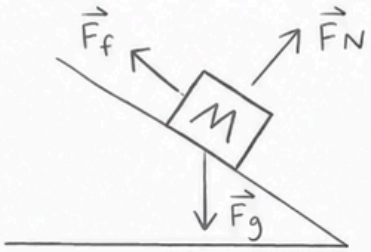
Recall: Free body diagrams shows the object free of its surroundings. The only forces acting **ON** the object are shown. The forces are drawn from the center of the object's mass.

Example 1) Draw a free- body diagram for the following scenarios:

a) A skydiver is descending with a constant velocity (consider air resistance).

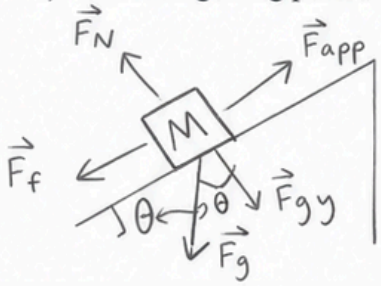


b) A toboggan sliding down the hill.



c) A block getting pulled up a ramp by a string.

$F_{gy} = F_g \cos\theta \qquad F_{gx} = F_g \sin\theta$



$\vec{F}_{app} = \vec{F}_T$ ↳ tension

Objects in Equilibrium

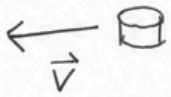
- **Equilibrium:** when the net force on an object is zero ($\sum\vec{F} = 0\text{N}$ **OR** both $\sum F_x = \sum F_y = 0\text{N}$)
- When you are working with force analysis, you must ALWAYS:
  1) Draw the free- body diagram.
  2) Consider the x & y components-choose and label which direction is (+) and which direction is (-) for EACH component.
  3) Draw each of the x and y components of the force vector on the FBD.
  4) Only once you have done this can you then use Newton's Law's to solve the problem.

① $\sum F = \vec{F}_1 + F_2 + \ldots$ ② $\sum F = m\vec{a}$

Kinds of Equilibrium

- **Static Equilibrium:** The simplest case of static equilibrium is when two forces are equal in magnitude but opposite in direction.
  - Object is at rest.
- **Dynamic Equilibrium:** Happens when bodies are moving at a constant velocity.
  - Still results in the sum of the forces acting on them being zero.

puck sliding on a frictionless surface of ice is in dynamic equilibrium



Question: Why would both situations be considered "equilibrium"?

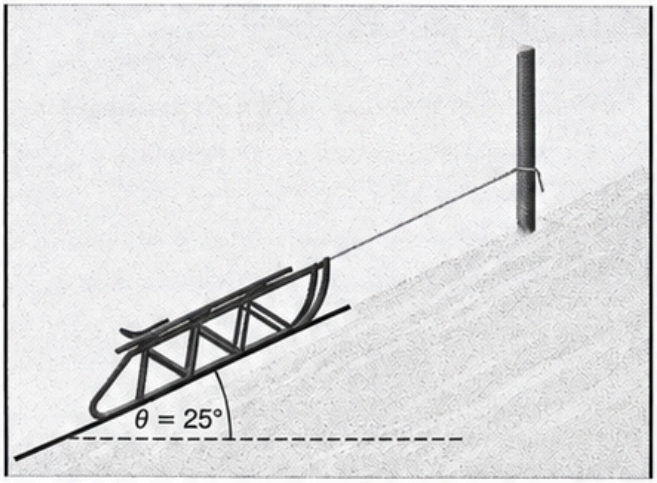
- **Static Equilibrium:** results in no net force, and no acceleration.
- **Dynamic Equilibrium:** also results in no net force.

Accelerating Objects

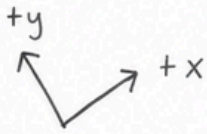
- If **acceleration** is said to be involved, then the object is **NOT in equilibrium**.
- In this case, you can use Newton's second law $\sum \vec{F} = m\vec{a}$
  - When solving these types of problems, set the positive x-axis in the direction of the net force (acceleration).
  - This will make the net force have no additional y-component, which will simplify the solution.
  - If you do not know the direction of the net force, then just set the positive x-axis in the direction that is more convenient to solve the problem.

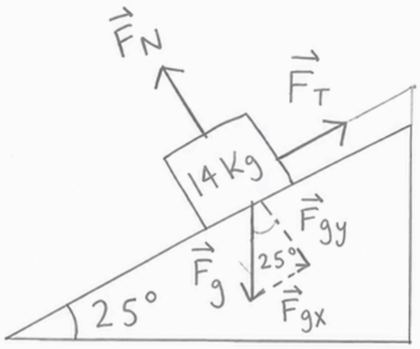
Equilibrium- Classroom Examples

Example 1: A sled has a mass of **14 kg** and is on a hill that is inclined **25°** to the horizontal, as shown in the figure below. The hill is very icy (friction is negligible), and the sled is held at rest by a rope attached to the post. The rope is parallel to the hill.



(a) Draw the FBD diagram of the scenario.





(b) Calculate the magnitude of the tension in the rope.

$\vec{F}_{gx} = -\sin 25° \times 137.34\ N$

$\vec{F}_{gx} = -58\ N$

$\vec{F}_{net\,x} = \vec{F}_{gx} + \vec{F}_T$

$\vec{F}_{net\,x} = m a_x$

$\vec{F}_{net\,x} = 0$

$0 = \vec{F}_{gx} + \vec{F}_T$

$0 = -58\ N + \vec{F}_T$

$\vec{F}_T = +58\ N$

$\vec{F}_g = mg$

$\vec{F}_g = (14)(9.81)$

$\vec{F}_g = 137.34\ N$

(c) Calculate the magnitude of the normal force acting on the sled.

$\vec{F}_{gy} = \cos 25° \times -137.34\ N$

$\vec{F}_{gy} = -124.5\ N$

$\vec{F}_{net_y} = \vec{F}_N + \vec{F}_{gy}$

$\vec{F}_{net_y} = ma_y$

$\vec{F}_{net_y} = 0$

$0 = \vec{F}_N + \vec{F}_{gy}$

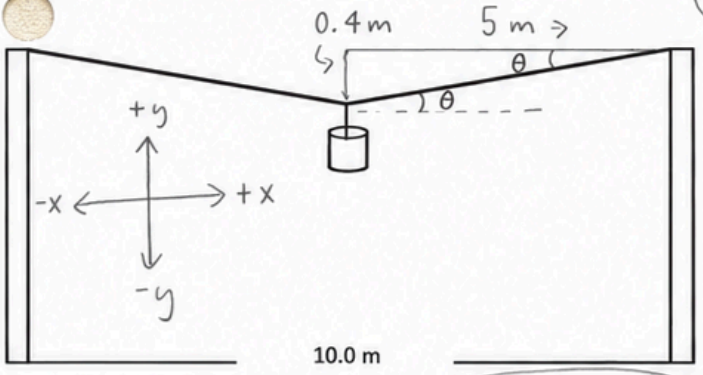
$0 = \vec{F}_N - 124.5\ N$

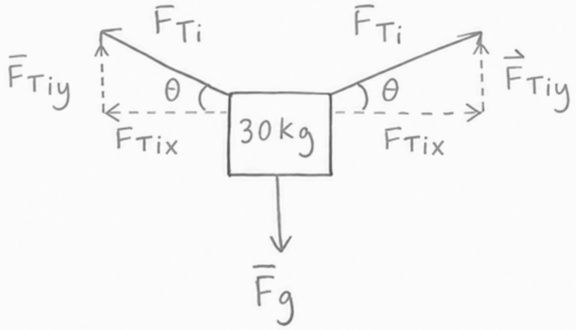
$\vec{F}_N = 124.5\ N$

$\vec{F}_N = +1.2 \times 10^2\ N$

<u>Example 2)</u> A clothesline is attached to high poles **10.0 m** apart. A pulley, allowed to roll freely on the line has a **30.kg** mass hanging from it. Find the magnitude of the tension in each half of the clothes-line if the sag at the centre is **0.40 m.** (Hint: Draw an FBD of the mass and resolve angled forces into components.) [1.8 x 10^3 N]

$F_T = 2F_{Ti}$ (sum of magnitudes) i = individual





no vectors, using only magnitudes

$\theta = \tan^{-1}(0.4/5)$

$\theta = 4.57°$

$F_g = 30 \times 9.81$

$\vec{F}_g = 294.3\ N$

$\vec{F}_{gy} = -294.3\ N$

$\vec{F}_{gx} = 0\ N$

$\vec{F}_{net_y} = F_{T_y} - F_g$

$F_{net_y} = ma_y$

$F_{net_y} = 0$

$0 = F_{T_y} - F_g$

$0 = F_{T_y} - 294.3\ N$

$F_{T_y} = 294.3\ N$

$F_{T_y} = \sin 4.57° \times F_T$

$294.3 = \sin 4.57° \times F_T$

$\frac{294.3}{\sin 4.57°} = \vec{F}_T$

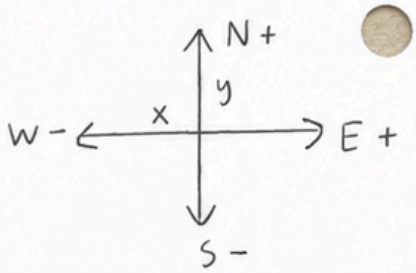
$3693.7\ N = F_T$

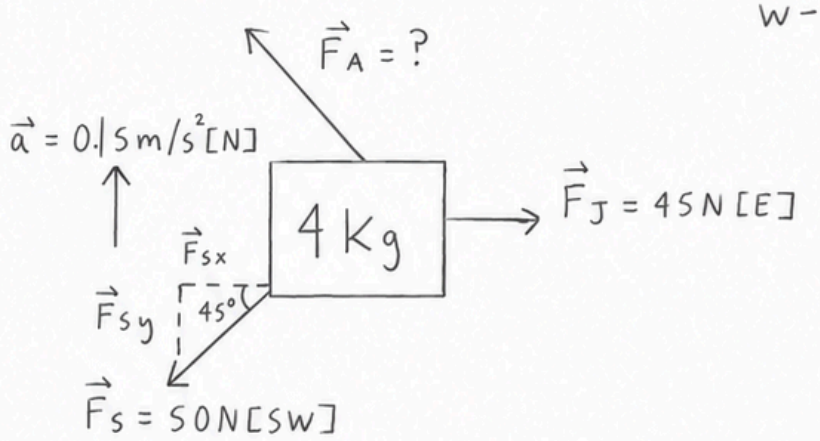
Sum of magnitudes; divide by 2 to find tension in each rope

$\vec{F}_{Ti} = 1846.85\ N$

$F_{Ti} = 1.8 \times 10^3\ N$

Example 3) Three students pull **horizontally** on ropes tied to a **4.0 *kg*** chair, causing it to accelerate at **0.50** $\frac{m}{s^2}$ **[*north*]**. If John pulls with **45 *N*** of force **[*E*]** and Saba pulls with **50. *N*** of force **[*SW*]**, what force is Aleksey exerting? Include an FBD with clearly labelled components.





$\vec{F}_{net\,x} = \vec{F}_{Jx} + \vec{F}_{Sx} + \vec{F}_{Ax}$

$\vec{F}_{net\,x} = m a_x$

$\vec{F}_{net\,x} = 0$

$0 = \vec{F}_{Jx} + \vec{F}_{Sx} + \vec{F}_{Ax}$

$0 = 45N - 35.36N + \vec{F}_{Ax}$

$\vec{F}_{Ax} = -9.64N$

$\vec{F}_{Jx} = +45N$ $\quad \vec{F}_{Jy} = 0N$

$\vec{F}_{Sx} = -\cos 45° \times \vec{F}_S$

$\vec{F}_{Sx} = -\cos 45° \times 50N$

$\vec{F}_{Sx} = -35.36N$

$\vec{F}_{Sy} = -\sin 45° \times \vec{F}_S$

$\vec{F}_{Sy} = -\sin 45° \times 50N$

$\vec{F}_{Sy} = -35.36N$

$\vec{F}_{net\,y} = \vec{F}_{Jy} + \vec{F}_{Sy} + \vec{F}_{Ay}$

$\vec{F}_{net\,y} = (4)(0.5)$

$\vec{F}_{net\,y} = 2N$

$2N = \vec{F}_{Jy} + \vec{F}_{Sy} + \vec{F}_{Ay}$

$2N = 0 - 35.36N + \vec{F}_{Ay}$

$\vec{F}_{Ay} = 2N + 35.36N$

$\vec{F}_{Ay} = 37.36N$

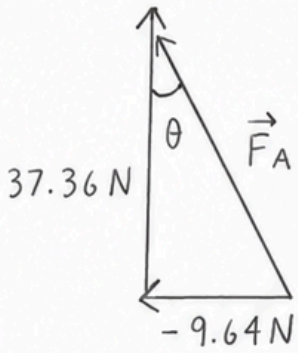
$\vec{F}_A = \sqrt{9.64^2 + 37.36^2}$

$\vec{F}_A = 38.6N$

$\theta = \tan^{-1}(9.64/37.36)$

$\theta = 14.47°$ west of north

$\therefore \vec{F}_A = 39N\ [N\,14°W]$



Homework

Pg. 65 #1-3, Pg. 69 #2, 6, 7, 8, 9